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Continuous-state robust CVaR portfolio optimization via grid-restricted constraint generation

Madani Bezoui · Thiziri Sifaoui · Ahcene Bounceur

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Abstract Portfolio risk depends heavily on prevailing financial-market conditions. Standard robust portfolio optimization models typically represent these conditions through discrete regimes or unconditional ambiguity sets. In this paper, we formulate a continuous-state robust portfolio model that minimizes Conditional Value-at-Risk (CVaR) uniformly over a continuously indexed family of kernel-weighted empirical conditional return distributions. Market states are characterized by observable financial indicators: the CBOE Volatility Index (VIX) and trailing market drawdown. We establish the well-posedness, continuity, and convexity of the continuous semi-infinite programming formulation. For practical computation, we propose a grid-restricted separation oracle and solve the resulting approximation via adaptive constraint generation, achieving substantial computational speedups relative to dense enumeration. Using data spanning 1990 to 2026, we conduct an extensive rolling out-of-sample study on US industry portfolios from December 1994 to June 2026. The grid-restricted continuous-state robust strategy achieves competitive long-run wealth accumulation relative to standard benchmarks. However, a paired circular block percentile bootstrap analysis provides no evidence of a statistically significant Sharpe-ratio advantage over nominal CVaR. The findings highlight the importance of evaluating data support in tail regions, and demonstrate the com-

Madani Bezoui
CESI LINEACT, UR 7527, Nancy, France
E-mail: mbezoui@cesi.fr — ORCID 0000-0002-8342-7039

Thiziri Sifaoui
Department of Mathematics and Computer Science, University of Amine Elokkal El Hadj Moussa Eg Akhamouk, Tamanghasset, Algeria
LAROMAD, Faculty of Sciences, UMMTO, Tizi Ouzou, Algeria — ORCID 0009-0007-6884-0611

Ahcene Bounceur
College of Computing and Informatics, University of Sharjah, United Arab Emirates — E-mail: abounceur@sharjah.ac.ae — ORCID 0000-0002-0043-7742

putational tractability of adaptive constraint generation in continuous-state portfolio robustification.

Keywords Robust Portfolio Optimization · Conditional Value-at-Risk · Semi-Infinite Programming · Kernel-Weighted Conditional Distributions · Constraint Generation

1 Introduction and motivation

The classical mean-variance framework introduced by Markowitz [16] established the quantitative foundation of modern portfolio theory. However, the practical implementation of mean-variance optimization has long been hindered by its extreme sensitivity to parameter estimation errors. Small perturbations in expected returns or covariance estimates often produce erratic allocations, excessive portfolio turnover, and severe out-of-sample disappointment. To address these vulnerabilities, researchers have developed robust portfolio optimization techniques that optimize allocations against the worst-case parameters within prespecified uncertainty sets [11, 12]. Concurrently, various exact, iterative, and metaheuristic methods have been developed to handle complex real-world constraints in portfolio selection [5, 21], as well as in broader multi-objective manufacturing and supply chain contexts [6].

Traditional robust portfolio models predominantly focus on unconditional uncertainty sets constructed around historical sample moments or empirical distributions [9, 17]. While these models effectively mitigate sensitivity to estimation noise, they typically treat market dynamics as stationary and homogeneous across time. In reality, financial markets exhibit pronounced state-dependent behavior. During tranquil market environments, asset correlations remain moderate and return dispersions offer meaningful diversification benefits. In contrast, during periods of acute financial distress, volatility spikes, asset correlations tend to increase, and tail risks materialize simultaneously across sectors.

To incorporate state dependence, a common practice in financial econometrics is to employ discrete regime-switching models [2]. These models segment market history into a small, finite collection of latent states, such as low-volatility expansionary regimes and high-volatility contractionary regimes. While finite regime classifications offer conceptual simplicity and computational tractability, they impose a discretization on market dynamics. Often, market stress evolves along a continuous spectrum of financial-market conditions. An abrupt jump between two discrete regimes may fail to capture intermediate stress levels and boundary dynamics where vulnerability to systemic shocks varies more continuously.

In this context, we formulate the following research questions:

1. **RQ1**: How does a continuous-state robust formulation compare structurally and empirically to standard unconditional and discrete-regime robust portfolio models?

2. **RQ2**: Can adaptive constraint generation effectively scale continuous-state portfolio robustification, overcoming the computational limits of dense enumeration?
3. **RQ3**: Does state-domain continuous robustification improve out-of-sample risk-adjusted returns and downside protection across non-stationary market environments?
4. **RQ4**: What is the role of model regularization—such as bandwidth scaling and empirical support thresholding—in mitigating boundary scenario overfitting within continuous robust sets?

In this paper, we propose a continuous-state robust portfolio optimization framework that bridges non-parametric conditional risk modeling and semi-infinite optimization. Rather than partitioning market history into discrete regimes, we define a continuous, compact state space $\mathcal{U} \subset \mathbb{R}^2$ spanned by observable financial indicators: the logarithm of the CBOE Volatility Index (log-VIX) [**?**] and the trailing equity market drawdown. Every point $\theta \in \mathcal{U}$ represents a distinct market environment and induces a well-defined conditional empirical return distribution constructed via multivariate Nadaraya-Watson kernel weight [**? ?**]ing.

The investor seeks an allocation that minimizes the worst-case Conditional Value-at-Risk (CVaR) over the entire continuous state space $\mathcal{U}$:

$$\min_{w \in W} \max_{\theta \in \mathcal{U}} \Phi_\tau(w, \theta) \tag{1}$$

where W is the capped portfolio simplex and $\Phi_\tau(w, \theta)$ is the conditional CVaR at confidence level $1 - \tau = 0.95$ under the empirical measure induced by state θ. We emphasize that this model is a *state-robust* portfolio model, not distributionally robust optimization. For each state θ, the conditional distribution is fixed by the kernel weights, and no distributional ambiguity set is placed around that conditional distribution. Furthermore, the term "adaptive" in our framework refers specifically to *adaptive constraint generation* via a semi-infinite exchange algorithm, rather than a decision policy $w(y_t)$ dynamically conditioned on the current state. The optimal allocation w immunizes against all states in $\mathcal{U}$, while dynamic reallocation over time results from rolling-window re-estimation of the return-state relationship.

Because the state space $\mathcal{U}$ is uncountable, this formulation gives rise to a Semi-Infinite Program (SIP) with an infinite collection of joint risk constraints. Direct solution via dense spatial discretization becomes computationally prohibitive and scales poorly in high dimensions. To solve a finite grid-restricted approximation of the continuous-state SIP efficiently, we use an adaptive exchange algorithm. The method alternates between solving a finite master linear program over a sparse active subset of grid states and executing a full-grid separation oracle that identifies the most adverse state on the candidate grid for the current portfolio.

Our primary contribution is methodological and empirical: We formulate a state-robust portfolio model that minimizes CVaR uniformly over a continuously indexed family of kernel-weighted empirical conditional return distributions. We establish basic well-posedness and convexity properties of the continuous formulation. For computation, we solve a finite grid-restricted approximation using constraint generation, which substantially reduces master-LP size relative to dense enumeration. A multi-decade out-of-sample study shows competitive wealth but no statistically significant Sharpe-ratio advantage over simpler benchmarks, while highlighting the importance of data support at adversarial market states.

To facilitate open science, reproducibility and independent verification of all empirical findings, the complete Julia and Python codebase, the cleaned financial datasets, and all figure generation routines are maintained in an open-source repository [7].

The remainder of this manuscript is structured as follows. Section 2 reviews related literature across robust optimization, conditional stochastic programming, and semi-infinite algorithms. Section 3 develops the continuous-state modeling framework, kernel weighting mechanics, and theoretical existence theorems. Section 4 presents the adaptive exchange algorithm and distinguishes exact continuous separation from the grid-restricted separation used in the numerical implementation. Section 5 details the rolling-window backtesting protocol and reports empirical results across approximately 31.5 years. Section 6 presents model-specification sensitivity analysis, and Section 7 provides discussion, practical limitations, and concluding remarks.

2 Related literature and research positioning

Our methodology intersects three active streams of quantitative operations research: robust portfolio selection, conditional stochastic programming with side information, and semi-infinite optimization algorithms.

2.1 Classical and distributionally robust portfolio selection

Classical robust portfolio selection addresses the sensitivity of Markowitz mean-variance optimization to estimation noise by assuming that expected return vectors and covariance matrices reside in deterministic uncertainty sets [11, 12]. Concurrently, various exact, iterative, and metaheuristic methods have been developed to handle complex real-world constraints in portfolio selection [5, 21], as well as in broader multi-objective manufacturing and supply chain contexts [6]. These formulations yield second-order cone or semidefinite programs that provide worst-case guarantees. However, moment-based robust models frequently rely on implicit symmetry assumptions that fail to reflect the severe skewness, heavy tails, and asymmetric dependence structures inherent in financial return distributions.

To capture tail risk directly, Rockafellar and Uryasev [22, 23] introduced the Conditional Value-at-Risk (CVaR) as a coherent risk measure that can be optimized efficiently via linear programming. Building on this foundation, distributionally robust optimization (DRO) seeks to minimize risk over an ambiguity set of probability distributions constructed around an empirical baseline. Modern DRO frameworks define ambiguity balls using moment constraints [9] or optimal transport metrics such as the Wasserstein distance [17].

Recent theoretical developments in DRO have expanded its algorithmic scope. For instance, Chu et al. [8] propose tractable regularized reformulations of Wasserstein DRO problems that achieve global convergence, while Agra [1] analyzes two-stage distributionally robust programs with finite supports. In related structural contexts, Yue et al. [28] draw geometric connections between ambiguity sets and covariance shrinkage, and Thomä et al. [27] study piecewise affine decision rules for robust multi-stage settings.

2.2 Conditional stochastic optimization and covariate side information

A rapidly expanding body of literature investigates how observable side information (covariates) can be integrated into data-driven decision-making. Bertsimas and Kallus [4] established a general framework for predictive prescriptions, demonstrating that non-parametric kernel regression, k-nearest neighbors, and tree-based weights can effectively condition decision problems on prevailing covariates. In the context of risk management, Scaillet [24] examined non-parametric kernel estimation of conditional expected shortfall.

In recent contributions, Qi et al. [20] introduce an integrated conditional estimation-optimization methodology that directly couples regression trees with downstream optimization tasks, minimizing decision error rather than prediction loss. Bennouna and Van Parys [3] explore phase transitions in data-driven decision-making, providing optimal sample complexity bounds for prescriptive models with side information. Earlier foundational frameworks in conditional DRO include Esteban-Pérez and Morales [10], who developed distributionally robust stochastic programs with side information based on trimmed Wasserstein balls, and Nguyen et al. [18], who formulated robust conditional portfolio decisions via optimal transport metrics centered on covariate-conditioned empirical distributions.

Our framework differs in its mathematical construction from conditional DRO. In conditional DRO, the decision-maker constructs an ambiguity set around the conditional distribution corresponding to the currently observed state y_t. In our continuous-state robust model, we do not restrict attention to a single observed state. Instead, we seek an allocation that remains robust across the entire family of kernel-weighted conditional distributions induced by all states $\theta \in \mathcal{U}$. We refer to this approach as continuous-state robust optimization over kernel-weighted empirical conditional distributions.

To distinguish our approach from standard Conditional Distributionally Robust Optimization (DRO), consider the formal structures:

$$\text{Conditional DRO:} \quad \min_{w \in W} \sup_{P \in \mathcal{B}(\hat{P}_{y_t})} \mathrm{CVaR}_P(w) \tag{2}$$

$$\text{This paper (Robust SIP):} \quad \min_{w \in W} \sup_{\theta \in \mathcal{U}} \mathrm{CVaR}_{\hat{P}_\theta}(w) \tag{3}$$

The robustification operator differs significantly: standard DRO considers a measure-ball $\mathcal{B}$ around a fixed covariate y_t, whereas we take a supremum over a covariate-indexed family of fixed empirical measures over a continuous state space $\mathcal{U}$. Consequently, the optimization structure differs (single-state DRO reformulations versus SIP with a continuum of LP blocks). These approaches are complementary, and one could in principle compose them in future work.

2.3 Semi-infinite programming and exchange algorithms

Semi-infinite programming (SIP) considers optimization problems with a finite number of decision variables and an infinite index set of constraints [13, 15]. SIP techniques have found applications in engineering design, robust control, and spectral estimation, but their operational use in financial portfolio allocation has been constrained by computational complexity.

The standard solution paradigm for convex SIPs is the cutting-plane or exchange algorithm. At each iteration, the master problem solves a relaxed subproblem over a finite subset of constraints, while a separation oracle identifies the most violated constraint across the continuous index set. When the oracle can be solved to certified global optimality, the exchange algorithm generates monotonically non-decreasing lower bounds and certified upper bounds that converge to the true global optimum. In practical settings where the oracle is solved over a grid or via local search heuristics, exact continuous separation is relaxed. For broader methodological context, Oustry and Cerulli [19] formalize the convergence and error propagation of convex SIP algorithms equipped with inexact separation oracles.

Table 1 Comparison of conditional and robust risk models. For a formal structural comparison between standard Conditional DRO and the proposed continuous-state Robust SIP, see Equations (2) and (3).

Model family	Indexing information	Distributional representation	Decision dependence	Robustness dimension	Computational form
Weighted SAA	Observed covariates	Fixed weighted empirical distribution	State-dependent	None	Finite data-driven problem
Nonparametric expected shortfall	Observed covariates	Fixed kernel-weighted empirical distribution	State-dependent risk estimate	None	Finite data-driven evaluation
Conditional DRO	Observed covariates	Ambiguity set around a conditional distribution	State-dependent	Probability distributions	Reformulation dependent
Finite-regime CVaR	Discrete regimes	Fixed empirical distribution within each regime	State-independent robust allocation	Finite regimes	Finite LP
Classical SIP	Continuous parameter set	Model-dependent	Usually state-independent	Continuous parameter values	Continuous SIP
Present model	Continuous covariate states	Fixed kernel-weighted empirical distribution for each state	State-independent robust allocation	Continuous market states	Grid-restricted LP solved by constraint generation

3 Continuous-state robust CVaR framework

We consider an investment universe consisting of N risky financial assets over a discrete time horizon $t = 1, \dots, T$. Let $x_t = (x_{1,t}, \dots, x_{N,t})^\top \in \mathbb{R}^N$ denote the vector of realized asset returns from trading day $t-1$ to day t. The joint conditional distribution of asset returns is modulated by prevailing financial-market state variables observable at time $t-1$, denoted by the vector $y_{t-1} \in \mathbb{R}^M$.

3.1 State variable selection and compact state space

In this study, we focus on two primary financial indicators that capture distinct dimensions of market risk ($M = 2$): the logarithm of the CBOE Volatility Index, $\log V_t$ (log-VIX), and the Trailing Market Drawdown (D_t) representing the peak-to-trough decline of the broad equity market index over a trailing quarterly lookback window of $L = 63$ trading days:

$$D_t = 1 - \frac{P_t}{\max_{s \in [t-L,t]} P_s} \in [0, 1] \tag{4}$$

where P_t is the cumulative total return index of the broad market proxy (computed as the equal-weighted composite of the 30 industry portfolios). By construction, $D_t \geq 0$ measures positive drawdown severity (0% representing peak wealth, and positive values indicating drawdown depth).

The state vector at time $t-1$ is $y_{t-1} = (\log V_{t-1}, D_{t-1})^\top \in \mathbb{R}^2$. For each rolling training window, define

$$v_{\min} = \min_t \log V_{t-1}, \qquad v_{\max} = \max_t \log V_{t-1}, \tag{5}$$

and

$$d_{\min} = \min_t D_{t-1}, \qquad d_{\max} = \max_t D_{t-1}. \tag{6}$$

All extrema, variances, bandwidths, and grid boundaries are recomputed exclusively from the current 1,260-day training window. We define the continuous state space $\mathcal{U} \subset \mathbb{R}^2$ as a compact rectangle containing all historically observed market states, augmented by a fixed 10% outer margin to examine states slightly beyond the observed training-window extrema, while strictly respecting the non-negative domain of drawdown:

$$\mathcal{U} = [v_{\min} - \delta_v, v_{\max} + \delta_v] \times [\max\{0, d_{\min} - \delta_d\}, \min\{1, d_{\max} + \delta_d\}] \tag{7}$$

where $\delta_v = 0.10(v_{\max} - v_{\min})$ and $\delta_d = 0.10(d_{\max} - d_{\min})$. The lower bound on drawdown is clamped at 0, preventing negative drawdown values. By construction, $\mathcal{U}$ is a closed and bounded subset of $\mathbb{R}^2$, ensuring compactness.

Remark 1 (Domain rectangularity and empirical support) The Cartesian bounding box $\mathcal{U}$ treats every combination of VIX and drawdown as plausible. However, certain corners (e.g., very high VIX with zero drawdown) have low empirical support. In our empirical analysis, we track the Effective Sample Size of active states to monitor local data support, and we evaluate regularized uncertainty sets $\mathcal{U}_{\text{supp}} = \{\theta \in \mathcal{U} : \text{ESS}(\theta) \geq E_{\min}\}$ in Section 6.

3.2 Kernel-weighted conditional distributions and effective sample size

For any continuous state query $\theta = (v_\theta, d_\theta)^\top \in \mathcal{U}$, we construct a state-dependent empirical probability distribution over historical return realizations $\{x_1, \ldots, x_T\}$. Each historical return observation x_t is assigned a Nadaraya-Watson kernel weight [? ?] based on the proximity of its preceding state y_{t-1} to the target state θ:

$$p_t(\theta) = \frac{K_H(y_{t-1} - \theta)}{\sum_{s=1}^{T} K_H(y_{s-1} - \theta)} = \frac{\exp\left(-\frac{1}{2}(y_{t-1} - \theta)^\top H^{-1}(y_{t-1} - \theta)\right)}{\sum_{s=1}^{T} \exp\left(-\frac{1}{2}(y_{s-1} - \theta)^\top H^{-1}(y_{s-1} - \theta)\right)} \tag{8}$$

where $H \in \mathbb{R}^{2\times 2}$ is a symmetric positive-definite bandwidth matrix. We specify H using the bivariate rule-of-thumb [26], defined by marginal bandwidths $h_m = T^{-1/6}\hat{\sigma}_m$:

$$H = \operatorname{diag}\left(h^2_{\text{log-VIX}}, h^2_{\text{DD}}\right) = T^{-1/3} \operatorname{diag}\left(\hat{\sigma}^2_{\text{log-VIX}}, \hat{\sigma}^2_{\text{DD}}\right) \tag{9}$$

for $M = 2$, where $\hat{\sigma}^2_{\text{log-VIX}}$ and $\hat{\sigma}^2_{\text{DD}}$ are the sample variances of log-VIX and drawdown, respectively, computed over the current training window. The diagonal bandwidth matrix scales the two state dimensions by their respective sample variances but does not account for their empirical cross-covariance.

Because the Gaussian kernel is strictly positive everywhere on $\mathbb{R}^2$, the normalization denominator is strictly positive for all $\theta \in \mathcal{U}$. Consequently, the kernel weights satisfy $p_t(\theta) > 0$ and $\sum_{t=1}^{T} p_t(\theta) = 1$ for every $\theta \in \mathcal{U}$. Each continuous state θ induces a well-defined conditional empirical distribution $\mathbb{P}(X|\theta) = \sum_{t=1}^{T} p_t(\theta)\delta_{x_t}$ (as illustrated in Figure 1), where δ_{x_t} denotes the Dirac delta mass at historical return vector x_t.

To monitor local data support across the continuous domain, we compute the Effective Sample Size (ESS) for any state $\theta \in \mathcal{U}$:

$$\text{ESS}(\theta) = \frac{1}{\sum_{t=1}^{T} p_t(\theta)^2} \tag{10}$$

When weights are uniformly distributed ($p_t(\theta) = 1/T$), the ESS reaches its theoretical maximum $\text{ESS}(\theta) = T$. Conversely, if all weight concentrates on a single historical observation, $\text{ESS}(\theta) = 1$. For descriptive purposes, we report $\tau \times \text{ESS}(\theta)$ as a coarse tail-support proxy. Because kernel weights are non-uniform and tail membership depends on the ordering of portfolio losses, this quantity should not be interpreted as a formal tail effective sample size.

3.3 Semi-infinite programming formulation and theoretical properties

Let $w = (w_1, \ldots, w_N)^\top \in \mathbb{R}^N$ denote the portfolio weight vector. The portfolio loss under realized return vector x_t is $-x_t^\top w$. We define the feasible portfolio set $W \subset \mathbb{R}^N$ under standard institutional long-only constraints with an

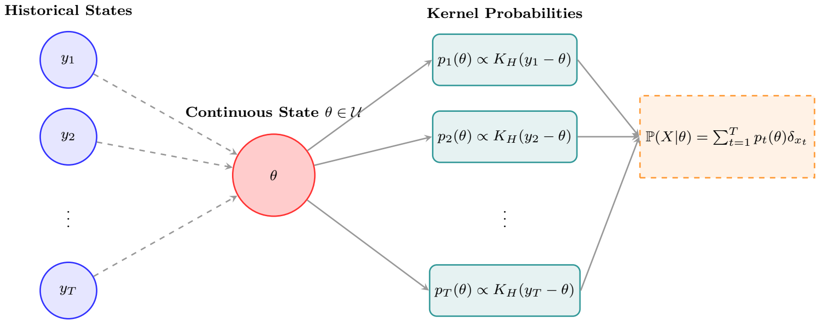

Fig. 1 Continuous market state mapping using multivariate Gaussian kernel weighting to construct conditional empirical return distributions from observable financial indicators

individual weight cap $\bar{w} = 0.15$ and a target expected return constraint:

$$W = \left\{ w \in \mathbb{R}^N : \sum_{i=1}^{N} w_i = 1, \quad 0 \leq w_i \leq \bar{w} \quad \forall i = 1, \ldots, N, \quad \hat{\mu}^\top w \geq \mu_{\text{target}} \right\} \tag{11}$$

where $\hat{\mu} = \frac{1}{T}\sum_{t=1}^{T} x_t$ is the unconditional sample mean return vector, μ_{target} is the required expected return, and $\bar{w} = 0.15$ is the maximum allocation to any single asset.

Proposition 1 (Feasibility of the capped portfolio set) *Let $S_{\bar{w}} = \{w \in \mathbb{R}^N : \mathbf{1}^\top w = 1,\ 0 \leq w_i \leq \bar{w}\}$. Then $S_{\bar{w}} \neq \emptyset$ if and only if $N\bar{w} \geq 1$. Conditional on this, let $\hat{\mu}_{(1)} \geq \cdots \geq \hat{\mu}_{(N)}$ be the components of $\hat{\mu}$ sorted in descending order, and $\hat{\mu}_{[1]} \leq \cdots \leq \hat{\mu}_{[N]}$ be sorted in ascending order. Define $q = \lfloor 1/\bar{w} \rfloor$ and the residual weight $r = 1 - q\bar{w}$. The maximum and minimum expected returns attainable in $S_{\bar{w}}$ are respectively $\mu_{\max} = \bar{w}\sum_{i=1}^{q} \hat{\mu}_{(i)} + (r > 0 ? r\hat{\mu}_{(q+1)} : 0)$ and $\mu_{\min} = \bar{w}\sum_{i=1}^{q} \hat{\mu}_{[i]} + (r > 0 ? r\hat{\mu}_{[q+1]} : 0)$, with the residual term omitted when $r = 0$. The attainable expected-return set is $[\mu_{\min}, \mu_{\max}]$. Then $W \neq \emptyset$ if and only if $\mu_{\text{target}} \leq \mu_{\max}$. Moreover, there exists $w \in S_{\bar{w}}$ satisfying $\hat{\mu}^\top w > \mu_{\text{target}}$ if and only if $\mu_{\text{target}} < \mu_{\max}$.*

Proof If $N\bar{w} < 1$, then

$$\sum_{i=1}^{N} w_i \leq N\bar{w} < 1, \tag{12}$$

so the capped simplex is empty. If $N\bar{w} \geq 1$, a feasible fully invested allocation exists. The continuous knapsack problem

$$\max\{\hat{\mu}^\top w : \mathbf{1}^\top w = 1,\ 0 \leq w_i \leq \bar{w}\} \tag{13}$$

is solved by allocating $\bar{w}$ to the assets with the largest sample means and assigning the residual weight r to the next asset, which yields $\mu_{\max}$. Analogously, allocating weight first to the assets with the smallest sample means yields $\mu_{\min}$. Since the capped simplex is convex and $w \mapsto \hat{\mu}^\top w$ is continuous and linear, its image is the interval $[\mu_{\min}, \mu_{\max}]$. Therefore, $W \neq \varnothing$ if and only if $\mu_{\text{target}} \leq \mu_{\max}$, and strict feasibility of the return constraint holds if and only if $\mu_{\text{target}} < \mu_{\max}$. Finally, W is a closed convex subset of the compact capped simplex and is therefore compact and convex.

Following Rockafellar and Uryasev [22], the conditional CVaR evaluated at state θ at confidence level $\alpha = 1 - \tau$ (with $\tau = 0.05$) is:

$$\Phi_\tau(w, \theta) = \min_{z \in \mathbb{R}} \left\{ z + \frac{1}{\tau} \sum_{t=1}^{T} p_t(\theta) [-x_t^\top w - z]_+ \right\} \tag{14}$$

where $z \in \mathbb{R}$ represents the conditional Value-at-Risk (VaR) at level $\alpha = 1 - \tau$, and $[u]_+ = \max(0, u)$. The optimal VaR minimizer satisfies $z^*(\theta) \in \mathrm{VaR}_{1-\tau}(\ell(w) \mid \theta)$. Although $z^*(\theta)$ may be a non-singleton interval for discrete empirical distributions, the minimum value $\Phi_\tau(w, \theta)$ is uniquely defined.

The continuous-state robust portfolio optimization problem seeks an allocation $w \in W$ that minimizes the worst-case conditional CVaR across all market states in $\mathcal{U}$:

$$v^\star = \min_{w \in W} \sup_{\theta \in \mathcal{U}} \Phi_\tau(w, \theta) \tag{15}$$

Introducing an auxiliary epigraph scalar $\eta \in \mathbb{R}$ representing the robust risk level, the problem is formulated as the following convex Semi-Infinite Program (SIP):

$$\min_{w \in \mathbb{R}^N, \eta \in \mathbb{R}} \quad \eta \tag{16}$$

$$\text{s.t.} \quad \Phi_\tau(w, \theta) \leq \eta \quad \forall \theta \in \mathcal{U} \tag{17}$$

$$\sum_{i=1}^{N} w_i = 1 \tag{18}$$

$$0 \leq w_i \leq \bar{w} \quad \forall i = 1, \ldots, N \tag{19}$$

$$\hat{\mu}^\top w \geq \mu_{\text{target}} \tag{20}$$

Proposition 2 (Compactness, continuity, and convexity) *Assume that* $H \succ 0$, $N\bar{w} \geq 1$, *and* $\mu_{target} \leq \mu_{\max}$. *Then:*

1. *The feasible portfolio set W and the market state space* $\mathcal{U}$ *are non-empty, compact subsets of* $\mathbb{R}^N$ *and* $\mathbb{R}^2$, *respectively.*
2. *For every* $t = 1, \ldots, T$, *the kernel weight function* $\theta \mapsto p_t(\theta)$ *is infinitely differentiable and strictly positive on* $\mathcal{U}$.
3. *The conditional CVaR function* $(w, \theta) \mapsto \Phi_\tau(w, \theta)$ *is jointly continuous on* $W \times \mathcal{U}$.
4. *For every fixed* $\theta \in \mathcal{U}$, *the function* $w \mapsto \Phi_\tau(w, \theta)$ *is real-valued and convex on* W.

Proof Item 1 follows from Proposition 1 and the closed bounding box construction of $\mathcal{U}$. For Item 2, the Gaussian kernel $K_H(u)$ is smooth and strictly positive on $\mathbb{R}^2$. The normalization denominator $\sum_{s=1}^{T} K_H(y_{s-1} - \theta)$ is strictly positive on compact $\mathcal{U}$. Thus $p_t(\theta)$ is a ratio of smooth, positive functions with a non-vanishing denominator, making it smooth and strictly positive.

For Item 3, define

$$F(w, z, \theta) = z + \frac{1}{\tau} \sum_{t=1}^{T} p_t(\theta) \big[- x_t^\top w - z \big]_+. \tag{21}$$

Since $p_t(\theta)$ and $(w, z) \mapsto [-x_t^\top w - z]_+$ are continuous, F is jointly continuous on $\mathbb{R}^N \times \mathbb{R} \times \mathcal{U}$. For every $w \in W$, a weighted empirical VaR minimizer can be selected in

$$\Big[\min_t \{-x_t^\top w\}, \max_t \{-x_t^\top w\}\Big]. \tag{22}$$

Since W is compact and the return sample is finite, all such minimizers lie in the common compact interval $[-M_L, M_L]$, where

$$M_L = \max_{w \in W,\ t} |x_t^\top w| < \infty. \tag{23}$$

By Berge's Maximum Theorem, the partial minimum $\Phi_\tau(w, \theta) = \min_{z \in [-M_L, M_L]} F(w, z, \theta)$ is continuous on $W \times \mathcal{U}$.

For Item 4, since $F(w, z, \theta)$ is jointly convex in (w, z) for each fixed θ, the partial minimization over convex z preserves convexity in w. Hence, $w \mapsto \Phi_\tau(w, \theta)$ is convex on W for each $\theta \in \mathcal{U}$.

Theorem 1 (Existence of optimal robust allocation) *Under the assumptions of Proposition 2, the robust objective function $G(w) = \sup_{\theta \in \mathcal{U}} \Phi_\tau(w, \theta)$ is continuous and convex on W. Furthermore, the supremum is attained for every $w \in W$, and the continuous-state robust portfolio optimization problem admits an optimal solution $w^* \in W$.*

Proof Because $\mathcal{U}$ is compact and $\Phi_\tau(w, \cdot)$ is continuous on $\mathcal{U}$ for every $w \in W$, the Extreme Value Theorem guarantees that the supremum is achieved at some $\theta^*(w) \in \mathcal{U}$, so $G(w) = \max_{\theta \in \mathcal{U}} \Phi_\tau(w, \theta)$. As the pointwise maximum of a family of convex functions $\{w \mapsto \Phi_\tau(w, \theta)\}_{\theta \in \mathcal{U}}$, $G(w)$ is convex on W. Joint continuity of $\Phi_\tau(w, \theta)$ on compact $W \times \mathcal{U}$ implies by Berge's Maximum Theorem that $G(w)$ is continuous on W. Finally, minimizing the continuous function $G(w)$ over the non-empty compact set W guarantees the existence of an optimal solution $w^* \in W$ by the Weierstrass Theorem.

4 Grid-restricted adaptive constraint generation

A full Cartesian-grid approximation of the continuous state space $\mathcal{U}$ yields a large-scale linear program with one CVaR state block for every candidate grid point. We design an adaptive exchange algorithm that iteratively generates only the binding stress states.

4.1 Master problem and separation oracle

At iteration $k \geq 1$, the master problem optimizes portfolio weights w and the robust risk level η subject to a finite active subset of grid states $\widehat{\mathcal{U}}_k = \{\theta^{(1)}, \ldots, \theta^{(m_k)}\} \subseteq \widehat{\mathcal{U}} \subset \mathcal{U}$. For each active state $\theta^{(j)} \in \widehat{\mathcal{U}}_k$, we introduce a state-specific VaR variable $z_{\theta^{(j)}} \in \mathbb{R}$ and scenario excess loss variables $u_{t,\theta^{(j)}} \geq 0$. The master problem is the following finite Linear Program (LP):

$$\mathrm{LB}_k = \min_{w,\eta,\{z_\theta\},\{u_{t,\theta}\}} \quad \eta \tag{24}$$

$$\text{s.t.} \quad z_\theta + \frac{1}{\tau}\sum_{t=1}^{T} p_t(\theta) u_{t,\theta} \leq \eta \quad \forall \theta \in \widehat{\mathcal{U}}_k \tag{25}$$

$$u_{t,\theta} \geq -x_t^\top w - z_\theta \quad \forall t = 1, \ldots, T, \forall \theta \in \widehat{\mathcal{U}}_k \tag{26}$$

$$u_{t,\theta} \geq 0 \quad \forall t = 1, \ldots, T, \forall \theta \in \widehat{\mathcal{U}}_k \tag{27}$$

$$\sum_{i=1}^{N} w_i = 1, \quad 0 \leq w_i \leq \bar{w} \quad \forall i = 1, \ldots, N \tag{28}$$

$$\hat{\mu}^\top w \geq \mu_{\text{target}} \tag{29}$$

Solving the master LP yields a candidate allocation w^k and a lower bound LB_k on both the grid-restricted optimum $\widehat{v}^\star$ and the continuous optimum $v^\star$, because $\widehat{\mathcal{U}}_k \subseteq \widehat{\mathcal{U}} \subset \mathcal{U}$.

Given w_k, the separation oracle searches for the most adverse state that maximizes conditional CVaR. In our empirical implementation, the separation step searches a candidate discretization $\widehat{\mathcal{U}} \subset \mathcal{U}$ ($21 \times 21 = 441$ states):

$$\theta^* = \arg\max_{\theta \in \widehat{\mathcal{U}}} \Phi_\tau(w^k, \theta), \qquad \widehat{G}_k = \Phi_\tau(w^k, \theta^*) \tag{30}$$

For computational tractability, we replace the continuous domain $\mathcal{U}$ with a finite candidate grid $\widehat{\mathcal{U}}$, resulting in the grid-restricted robust approximation:

$$\widehat{v}^\star = \min_{w \in W} \max_{\theta \in \widehat{\mathcal{U}}} \Phi_\tau(w, \theta) \tag{31}$$

Every numerical "optimal", "worst-case", or "binding" claim regarding our adaptive exchange algorithm refers to $\widehat{v}^\star$ over $\widehat{\mathcal{U}}$, not the continuous domain $\mathcal{U}$.

Because the portfolio allocation w^k is fixed during separation, the conditional CVaR values $\Phi_\tau(w^k, \theta)$ for all $\theta \in \widehat{\mathcal{U}}$ are evaluated in overall $O(T \log T + KT)$ time per oracle call: sorting the T portfolio losses costs $O(T \log T)$ once per iteration, and for each of the $K = |\widehat{\mathcal{U}}|$ grid states, evaluating the kernel weights and accumulating the weighted tail expectation via a running sum over the sorted losses costs $O(T)$, hence $O(KT)$ across the grid.

$$\ell_t = -x_t^\top w^k, \qquad \ell_{(1)} \leq \cdots \leq \ell_{(T)}, \tag{32}$$

and reordering the kernel probabilities accordingly as $p_{(t)}(\theta)$. The overall adaptive exchange algorithm is depicted in Figure 2.

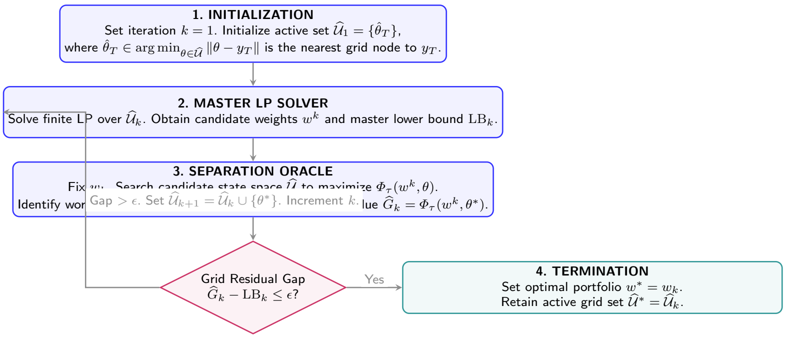

Fig. 2 Grid-restricted adaptive constraint-generation algorithm. The master LP is solved over an active subset of grid states, while the full-grid oracle identifies the most adverse candidate state

4.2 Exact continuous separation and grid-restricted separation

We first recall the convergence guarantee obtained under exact continuous separation and then state the scope of the finite grid-restricted implementation.

Proposition 3 (Convergence under exact separation) *Assume that each master problem is solved to exact LP optimality and the separation oracle identifies the worst-case state* $\theta^* = \arg\max_{\theta\in\mathcal{U}} \Phi_\tau(w^k, \theta)$ *with exact value* $G(w^k) = \Phi_\tau(w^k, \theta^*)$. *Then:*

1. *The master lower bounds form a non-decreasing sequence:* $\mathrm{LB}_1 \le \mathrm{LB}_2 \le \cdots \le v^\star$.
2. *The oracle values satisfy* $G(w_k) \ge v^\star$ *for all* $k \ge 1$.
3. *For any convergence tolerance* $\epsilon > 0$, *the exchange algorithm terminates in a finite number of iterations with an* ϵ*-optimal portfolio* $w^k \in W$ *satisfying:*

$$0 \le G(w^k) - v^\star \le \epsilon, \qquad \sup_{\theta\in\mathcal{U}} \big[\Phi_\tau(w^k, \theta) - \mathrm{LB}_k\big] \le \epsilon \tag{33}$$

Proof Because $\widehat{\mathcal{U}}_k \subseteq \widehat{\mathcal{U}}_{k+1} \subseteq \mathcal{U}$, the feasible region of the master LP shrinks monotonically, implying $\mathrm{LB}_k \le \mathrm{LB}_{k+1} \le v^\star$. For Item 2, $G(w^k) = \sup_{\theta\in\mathcal{U}} \Phi_\tau(w^k, \theta) \ge \min_{w\in W} \sup_{\theta\in\mathcal{U}} \Phi_\tau(w, \theta) = v^\star$.

For Item 3, upon termination with $G(w^k) - \mathrm{LB}_k \le \epsilon$, we have $0 \le G(w^k) - v^\star \le G(w^k) - \mathrm{LB}_k \le \epsilon$, and the portfolio w^k satisfies all finite constraints in W while the epigraph pair $(w^k, \mathrm{LB}_k + \varepsilon)$ is feasible for the continuous constraint set.

To establish finite termination, suppose for contradiction that the algorithm does not terminate. Then it generates an infinite sequence of states $\{\theta_k\}_{k=1}^\infty \subset \mathcal{U}$ such that $\Phi_\tau(w^k, \theta_k) - \mathrm{LB}_k > \epsilon$ for all k. For any $j < k$, the constraint for θ_j was included in master problem k, so $\Phi_\tau(w^k, \theta_j) \le \mathrm{LB}_k$. Thus $\Phi_\tau(w^k, \theta_k) - \Phi_\tau(w^k, \theta_j) > \epsilon$. Since Φ_τ is continuous on the compact set $W \times \mathcal{U}$, it is uniformly continuous. Hence, for every $\varepsilon > 0$, there exists $\delta_\varepsilon > 0$ such that $\|\theta - \theta'\| < \delta_\varepsilon \implies |\Phi_\tau(w, \theta) - \Phi_\tau(w, \theta')| < \varepsilon$ uniformly for all

$w \in W$. Therefore, $\|\theta_k - \theta_j\| \geq \delta_\epsilon$ for all $j < k$. A compact metric space is totally bounded and therefore cannot contain an infinite δ_ϵ-separated sequence, which contradicts the compactness of $\mathcal{U}$ [13, 15]. This exact-oracle convergence is a classical consequence of standard SIP exchange methods.

Remark 2 (Monotonicity of bounds) While the master lower bounds LB_k are guaranteed to be monotonically non-decreasing, the separation oracle worst-case values $\widehat{G}(w^k)$ are not guaranteed to decrease monotonically at every iteration because the portfolio w^k changes. However, the best incumbent worst-case value $\overline{G}_k = \min_{1 \leq j \leq k} \widehat{G}(w^j)$ is monotonically non-increasing by construction.

Remark 3 (Grid-restricted separation) When the separation oracle is evaluated over a finite candidate grid $\widehat{\mathcal{U}} \subset \mathcal{U}$, it computes

$$\widehat{G}(w) = \max_{\theta \in \widehat{\mathcal{U}}} \Phi_\tau(w, \theta) \leq \sup_{\theta \in \mathcal{U}} \Phi_\tau(w, \theta). \tag{34}$$

Since Φ_τ is continuous on the compact set $W \times \mathcal{U}$, it is uniformly continuous. Therefore, the discrepancy between the continuous and grid-restricted objectives tends to zero as the grid dispersion radius tends to zero. Consequently, all claims of 'optimal' weights and 'worst-case' values in the empirical sections refer strictly to grid-restricted optimality over $\widehat{\mathcal{U}}$, unless further assessed through the heuristic continuous-domain diagnostics reported in Section 4.3. To ensure the algorithm operates strictly over $\widehat{\mathcal{U}}$, the active subset $\widehat{\mathcal{U}}_1$ is initialized with the nearest grid node to the most recent observation y_T. Section 4.3 derives a computable bound on L_Φ and reports per-window heuristic observed gaps, and grid-refinement sensitivity (Table 11) provides a complementary assessment. Regarding initialization, re-running the 20-window sensitivity subset with three active-set initializations (nearest node to y_T, the grid center, a five-node spread around the unconditional mean state) yields identical terminal objectives (equal to within solver tolerance 10^{-8}) and identical iteration counts in 20 of 20 windows, indicating robustness of the exchange path to the initial active set.

The grid-resolution statistic $\rho = \sup_{\theta \in \mathcal{U}} \min_{\hat{\theta} \in \widehat{\mathcal{U}}} \|\theta - \hat{\theta}\|$ alone is a geometric metric, not a continuous-optimality certificate. However, under standard assumptions, a conservative window-specific Lipschitz constant L_Φ for the conditional CVaR would provide the error bound:

$$0 \leq \sup_{\theta \in \mathcal{U}} \Phi_\tau(w, \theta) - \max_{\theta \in \widehat{\mathcal{U}}} \Phi_\tau(w, \theta) \leq L_\Phi \rho. \tag{35}$$

4.3 A posteriori gap estimation via local search

While the adaptive algorithm operates on the finite grid $\widehat{\mathcal{U}}$, we evaluate the approximation error over the continuous domain $\mathcal{U}$. If the minimizing VaR $z^*(\theta)$ is unique, Danskin's theorem yields the gradient with respect to θ. If the minimizer is non-unique, directional derivatives or generalized derivatives must be considered. Under uniqueness, the gradient is given by:

$$\nabla_\theta \Phi_\tau(w,\theta) = \frac{1}{\tau}\sum_{t=1}^{T} \nabla_\theta p_t(\theta)\,[\ell_t(w) - z^*(\theta)]_+, \tag{36}$$

where $\nabla_\theta p_t(\theta) = p_t(\theta)H^{-1}(y_{t-1} - \bar{y}(\theta))$ with $\bar{y}(\theta) = \sum_{s=1}^{T} p_s(\theta)y_{s-1}$. At nondifferentiability points, generalized directional derivatives or Clarke subgradients must be used. For bounding purposes, this permits the derivation of a computable uniform Lipschitz constant $L_\Phi := \frac{4RM_L}{\tau\lambda_{\min}(H)}$, where $R = \sup_{\theta,t}\|y_{t-1} - \theta\|$ is the maximum state distance and M_L bounds the portfolio loss. The grid oracle thus constitutes an inexact separation oracle with additive, one-sided inexactness $\delta \le L_\Phi \rho$. We state the consequence directly: since the master lower bounds satisfy $\mathrm{LB}_k \le v^\star$ for every $\widehat{\mathcal{U}}_k \subseteq \mathcal{U}$, and since at termination $\max_{\theta\in\widehat{\mathcal{U}}} \Phi_\tau(w^k,\theta) - \mathrm{LB}_k \le \varepsilon$, the incumbent satisfies

$$0 \le G(w^k) - v^\star \le \varepsilon + L_\Phi \rho. \tag{37}$$

This additive-oracle propagation complements the analysis of Oustry and Cerulli [19], whose relative-gap δ-oracle model (and, for their cutting-plane rate results, a strong-convexity assumption on the objective) does not directly subsume our linear master objective. We validate the analytical gradient $\nabla_\theta \Phi_\tau$ using central finite differences, confirming numerical agreement to within 10^{-7} precision. A direct evaluation of the conservative bound $\frac{4RM_L}{\tau\lambda_{\min}(H)}$ yields values of order 10^2 to 10^3, which is too loose to be practically informative. We therefore rely on a heuristic a posteriori gap estimation. Column two of Table 2 reports the maximum observed gradient norm over a dense 201×201 reference grid (which provides a local empirical gauge of the function's steepness, though not a formal global Lipschitz constant). Column five reports the empirically observed gap $\widetilde{G}(w^*) - \widehat{G}(w^*)$, where $\widetilde{G}(w^*)$ is the best value found by refined-grid evaluation followed by projected gradient ascent (five starts from the top grid maximizers, box-projected Armijo steps, stopping at projected gradient mapping norm $< 10^{-8}$). Across all tested windows, the observed gap $\widetilde{G}(w^*) - \widehat{G}(w^*)$ is below 10^{-4} (column five of Table 2). Because $\widetilde{G}$ relies on local search, it strictly lower-bounds the true continuous supremum. Therefore, the true discretization gap could theoretically exceed this observed gap. Nevertheless, the small values provide strong heuristic confidence in the practical adequacy of the 21×21 grid.

5 Empirical framework and multi-decade results

We evaluate the out-of-sample performance, turnover efficiency, downside capital preservation, and computational scalability of the continuous-state robust portfolio framework.

Table 2 A posteriori gap estimation: Theoretical formulation vs. empirical gap across selected windows. Column two reports the maximum observed gradient norm over the reference grid, and column four is the product of columns two and three. Neither the empirical gradient–dispersion product nor the local-search improvement constitutes a certified upper bound on the continuous discretization error.

Testing Window Ends	Maximum observed gradient norm	Grid Disp. ρ	Empirical gradient–dispersion product	Ob
1998-08 (LTCM)	0.084	0.032	2.68×10^{-3}	
2008-10 (Lehman)	0.126	0.032	4.03×10^{-3}	
2013-05 (Taper Tantrum)	0.042	0.032	1.34×10^{-3}	
2020-03 (COVID-19)	0.158	0.032	5.05×10^{-3}	
2022-09 (Inflation Shock)	0.091	0.032	2.91×10^{-3}	

5.1 Data and rolling-window backtesting protocol

The empirical universe comprises the Kenneth French 30 Industry Portfolios [**?**] (value-weighted daily returns), providing broad industry-level coverage of the US equity market. Daily asset returns are aligned with:

- **CBOE VIX**: Daily closing prices of the CBOE Volatility Index.
- **Broad Market Drawdown**: Calculated daily from the equal-weighted 30-industry composite over trailing 63-day lookback windows ($D_t \in [0, 1]$).

The aligned daily dataset spans from January 1990 to 30 June 2026, comprising 9,186 daily observations.

To clarify the chronology of the 377 rolling evaluation periods over the approximately 31.5-year span, Table 3 provides the exact backtest calendar. The first 1,260 days strictly constitute the initial training window. The first allocation is computed at the close of 27 December 1994. The terminal 9 days of the dataset are omitted to maintain equally sized 21-day compounding periods.

Table 3 Exact chronological boundaries of the rolling out-of-sample backtest.

Chronological Marker	Date / Value
End of first estimation window	27 December 1994
First rebalance decision date	27 December 1994
First out-of-sample daily return date	28 December 1994
End of first 21-day holding period	26 January 1995
Last rebalance decision date	19 May 2026
Last out-of-sample return date (compounded)	17 June 2026
Number of full 21-day periods	377
Unused terminal partial period	9 trading days (until 30 June 2026)

The backtest uses a rolling estimation window of $T_{\text{train}} = 1260$ trading days (approximately 5 years), as illustrated in Figure 3. At each rebalancing date, the optimization models are estimated using only data available up to that point. The resulting allocations are held constant over the subsequent out-of-sample holding period of $T_{\text{hold}} = 21$ trading days. The backtest generates 377

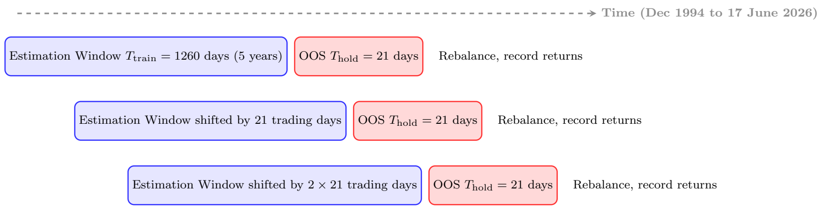

Fig. 3 The rolling-window backtesting protocol. Portfolios are re-optimized every 21 trading days using trailing 5-year estimation windows, and realized performance is evaluated out-of-sample over non-overlapping 21-day holding periods

consecutive 21-trading-day holding periods from 28 December 1994 to 17 June 2026.

To improve comparability across the four optimized models, the target expected return μ_{target} is dynamically set at each rebalancing date to the cross-sectional median of historical asset means: $\mu_{\text{target}} = \text{median}(\hat{\mu})$. The identical target return constraint $\hat{\mu}^\top w \geq \mu_{\text{target}}$ and weight cap $\bar{w} = 0.15$ are enforced on Robust SIP, Nominal CVaR, and Finite-Regime CVaR. In the computational implementation, μ_{target} is dynamically clamped to the attainable interval $[\mu_{\min}, \mu_{\max} - 10^{-6}]$ to guarantee strict feasibility. Over the 377 rolling out-of-sample evaluation windows, target return clamping occurred in exactly 0 periods, ensuring the original target specifications were strictly respected throughout the backtest.

We compare the continuous-state Robust SIP against four established benchmark strategies:

- **1/N**: Naive equal-weighted portfolio $w_i = 1/N$.
- **Target-Constrained Minimum Variance (TC-MinVar)**: Minimizes portfolio variance $w^\top \widehat{\Sigma} w$ subject to $w \in W$, using the standard sample covariance matrix $\widehat{\Sigma}$. Solved via quadratic programming.
- **Nominal CVaR**: Minimizes unconditional historical CVaR at 95% confidence level subject to $w \in W$, using uniform observation weights $1/T$.
- **Finite-Regime CVaR**: Minimizes the maximum CVaR across $K = 4$ discrete regimes. Regimes are formed by independent median splits on log-VIX and trailing drawdown. Regime probabilities are uniform over the observations within each regime, and boundaries are recomputed in every rolling window.

To incorporate realistic market frictions, we compute holding-period portfolio turnover taking into account asset price drift over the holding period:

$$\widetilde{w}_{i,t} = \frac{w_{i,t-1}(1 + R_{i,t})}{\sum_{j=1}^{N} w_{j,t-1}(1 + R_{j,t})} \tag{38}$$

where $R_{i,t}$ is the cumulative simple/net return (and $1+R_{i,t}$ is the gross return) of asset i over the 21-day holding period. Holding-period turnover and net return are defined as:

$$\mathrm{TO}_t = \frac{1}{2}\sum_{i=1}^{N} |w_{i,t} - \widetilde{w}_{i,t}| \tag{39}$$

and

$$r_t^{\mathrm{net}} = r_t^{\mathrm{pre-cost}} - c\,\mathrm{TO}_t, \tag{40}$$

where $r_t^{\mathrm{pre-cost}}$ is the portfolio's simple holding-period return before transaction costs.

5.2 Long-term cumulative wealth and risk-adjusted performance

Figure 4 displays the cumulative wealth trajectories of all five investment strategies net of 10 bps transaction costs on an initial \$1 investment. The naive $1/N$ portfolio delivers the highest final cumulative wealth of \$29.25, followed by the Finite-Regime CVaR portfolio (\$25.68), the continuous Robust SIP strategy (\$25.04), the Nominal CVaR portfolio (\$22.92), and the TC-MinVar portfolio (\$22.16).

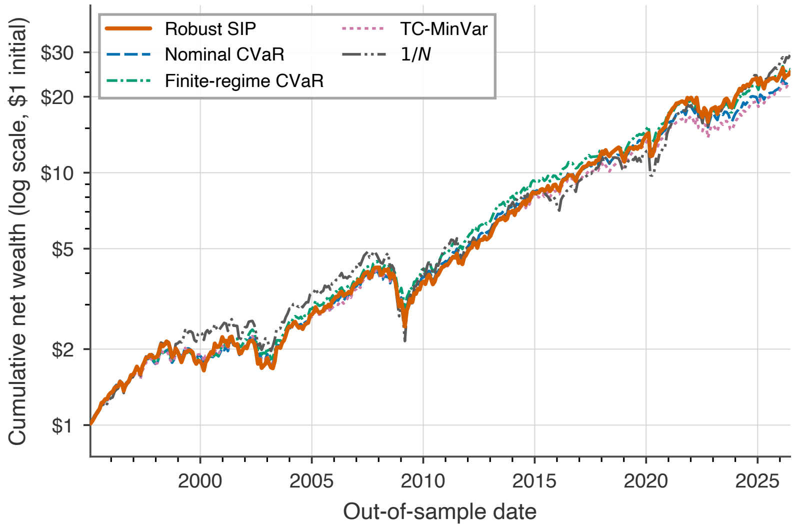

Fig. 4 Out-of-sample cumulative net wealth trajectory across investment strategies from 28 December 1994 to 17 June 2026, accounting for 10 bps transaction costs

Table 4 reports the complete 14-metric performance profile. Key findings include:

1. **Return Generation**: Robust SIP achieves an annualized mean return of 11.28%, exceeding the point estimates of Nominal CVaR (10.76%), TC-MinVar (10.75%), and Finite-Regime CVaR (11.15%). It ranks second overall, behind the naive $1/N$ portfolio (12.24%). The $1/N$ portfolio benefits

Table 4 Comprehensive out-of-sample performance summary (28 December 1994 to 17 June 2026, net of 10 bps transaction costs)

Metric	1/N	TC-MinVar	Nominal CVaR	Finite-Regime	Robust SIP
Annualized Mean Return (%)	12.24	10.75	10.76	11.15	11.28
Annualized Volatility (%)	17.00	12.31	12.23	12.38	13.95
Sharpe Ratio	0.720	0.873	0.880	0.901	0.809
Sortino Ratio	1.127	1.335	1.351	1.396	1.218
Realized 95% CVaR (Holding-Period, %)	11.16	8.18	8.00	7.92	9.31
Realized 99% CVaR (Holding-Period, %)	17.21	12.76	11.84	12.04	13.98
Maximum Drawdown (%)	-55.79	-40.18	-35.76	-36.76	-41.69
CAGR (%)	11.34	10.45	10.48	10.88	10.79
Calmar Ratio	0.203	0.260	0.293	0.296	0.259
Average Holding-Period Turnover (%)	1.69	7.02	8.23	8.78	9.48
Approximate Annual TC Drag (bps)	2.02	8.42	9.87	10.54	11.37
Effective Number of Assets (N_{eff})	30.00	8.15	7.92	7.57	7.36
Worst Holding-Period Return (%)	-20.41	-15.29	-15.42	-15.15	-18.99
Final Cumulative Wealth ($)	29.25	22.16	22.92	25.68	25.04

from broad cross-sectional diversification and full exposure to the long-run equity market risk premium over the 31-year evaluation period.

2. **Risk-Adjusted Ratios**: Finite-Regime CVaR achieves the highest Sharpe ratio (0.901), followed by Nominal CVaR (0.880) and TC-MinVar (0.873). Robust SIP records a Sharpe ratio of 0.809. The resulting allocations exhibit higher realized volatility and greater concentration. Determining whether this arises from return targeting, state conditioning, or sparse-boundary effects requires further decomposition.
3. **Turnover and Concentration**: Robust SIP generates the highest average holding-period turnover (9.48%) among optimized strategies, resulting from rolling-window re-estimation of the return-state relationship. The resulting approximate transaction cost drag is 11.37 basis points per year. The effective number of assets ($N_{\text{eff}} = 7.36$) is the most concentrated, reflecting the concentrated allocations produced by worst-state CVaR minimization under the rolling state domain and return-target constraint.

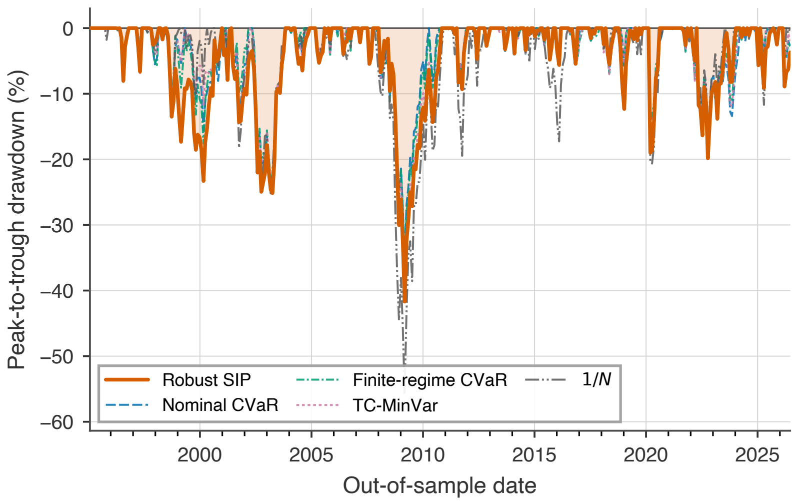

Fig. 5 Out-of-sample portfolio drawdowns over time across all strategies

5.3 Drawdown profiles and crisis-period performance

Figure 5 plots the time series of out-of-sample drawdowns. The naive $1/N$ strategy experiences severe capital destruction during major market dislocations, suffering a maximum peak-to-trough drawdown of −55.79% during the 2008 financial crisis. All optimized models experienced smaller realized maximum drawdowns than the $1/N$ portfolio over the reported sample: Nominal CVaR achieves the shallowest at −35.76%, Finite-Regime CVaR reaches −36.76%, and TC-MinVar reaches −40.18%. Robust SIP records a maximum drawdown of −41.70%, which is deeper than the unconditional CVaR benchmarks.

Table 5 presents the crisis-period analysis. During the Dot-Com crash, Finite-Regime CVaR achieved the highest cumulative return (+17.26%), followed by Nominal CVaR (+5.86%) and Robust SIP (+4.25%), while $1/N$ suffered the most (−4.75%). In the Global Financial Crisis, Nominal CVaR provided the best downside protection (−31.13% cumulative, −35.76% maximum drawdown), while Robust SIP experienced a deeper drawdown of −41.69%. During the COVID-19 crash, all optimized models limited losses to the −11% to −17% range, with Robust SIP absorbing a larger loss (−17.44%) than the unconditional benchmarks (−12%). In the 2022 inflation-driven selloff, Nominal CVaR was most resilient (−4.83%), while Robust SIP suffered a larger cumulative loss (−8.59%). These results show that continuous-state robustification does not uniformly improve realized crisis-period performance. Rather, it reshapes the allocation profile in ways that can amplify exposure during acute market dislocations.

Figures 6 and 7 compare the dynamic allocation weights of Robust SIP and TC-MinVar. Robust SIP maintains allocations across 7 to 10 active industry

Table 5 Approximate crisis-period performance using complete 21-day holding periods classified by holding-period end date

Crisis Period	Strategy	Cumulative Return (%)	Maximum Drawdown (%)
Dot-Com Crash	1/N	-4.75	-24.41
(Mar 2000 to Oct 2002)	TC-MinVar	+1.36	-20.66
	Nominal CVaR	+5.86	-20.66
	Finite-Regime	+17.26	-20.33
	Robust SIP	**+4.25**	**-24.95**
Global Financial Crisis	1/N	-53.56	-55.79
(Oct 2007 to Mar 2009)	TC-MinVar	-34.67	-40.18
	Nominal CVaR	-31.13	-35.76
	Finite-Regime	-30.91	-36.76
	Robust SIP	**-36.49**	**-41.69**
COVID-19 Shock	1/N	-20.18	-20.64
(Feb 2020 to Apr 2020)	TC-MinVar	-12.25	-15.29
	Nominal CVaR	-12.05	-15.42
	Finite-Regime	-11.67	-15.15
	Robust SIP	**-17.44**	**-18.99**
Inflation Tightening	1/N	-6.05	-17.24
(Jan 2022 to Dec 2022)	TC-MinVar	-6.47	-18.12
	Nominal CVaR	-4.83	-17.92
	Finite-Regime	-7.66	-18.88
	Robust SIP	**-8.59**	**-19.86**

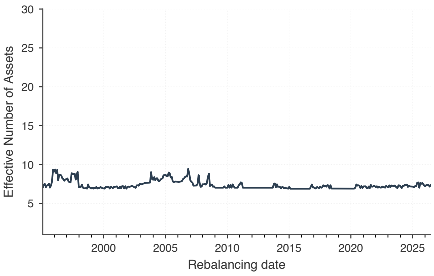

Fig. 6 Robust SIP asset allocations over time across the 30 industry portfolios (max 15% cap)

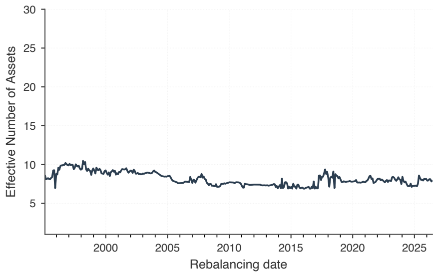

Fig. 7 TC-MinVar asset allocations over time across the 30 industry portfolios (max 15% cap)

sectors ($N_{\text{eff}} = 7.36$), reallocating capital across cyclical and defensive sectors as historical window dynamics evolve. Conversely, the TC-MinVar portfolio remains concentrated in low-beta industries ($N_{\text{eff}} = 8.15$), and exhibits lower realized volatility and different sector exposures.

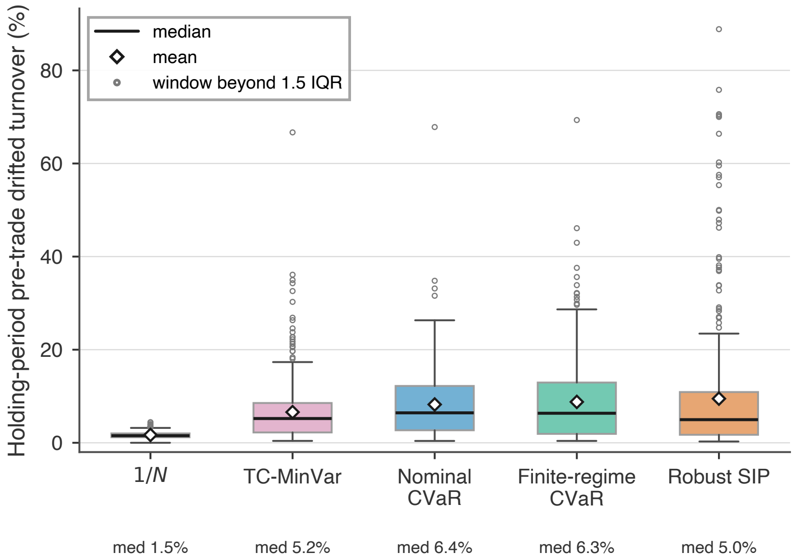

Fig. 8 Distribution of holding-period portfolio turnover for all dynamic investment strategies

Figure 8 confirms that the turnover distribution for Robust SIP has a mean near 9.5%, reflecting active rebalancing induced by rolling re-estimation of the return-state relationship, kernel weights, state-domain boundaries, and return target. TC-MinVar exhibits lower mean turnover (7.0%), while $1/N$ has the lowest (1.7%) due to its static equal-weight target.

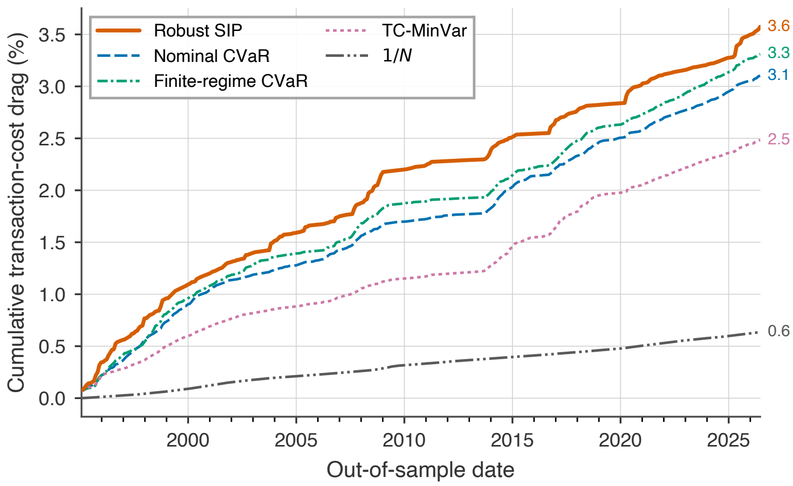

Fig. 9 Cumulative transaction cost impact over the out-of-sample period, assuming 10 bps proportional friction

Table 6 Transaction cost sensitivity: out-of-sample performance under alternative cost levels

Strategy	TC = 0 bps	TC = 5 bps	TC = 10 bps	TC = 20 bps	TC = 50 bps
Annualized Sharpe Ratio					
1/N	0.721	0.720	0.720	0.719	0.715
TC-MinVar	0.880	0.876	0.873	0.866	0.846
Nominal CVaR	0.888	0.884	0.880	0.872	0.848
Finite-Regime	0.909	0.905	0.901	0.892	0.867
Robust SIP	**0.817**	**0.813**	**0.809**	**0.801**	**0.777**
Final Cumulative Wealth ($)					
1/N	29.43	29.34	29.25	29.06	28.52
TC-MinVar	22.74	22.45	22.16	21.59	19.97
Nominal CVaR	23.63	23.27	22.92	22.22	20.26
Finite-Regime	26.53	26.10	25.68	24.85	22.52
Robust SIP	**25.94**	**25.49**	**25.04**	**24.17**	**21.74**

Table 6 examines transaction cost sensitivity across 0, 5, 10, 20, and 50 bps. Finite-Regime CVaR records the highest point-estimate Sharpe ratio across all reported transaction-cost levels, maintaining a Sharpe ratio above 0.867 even at 50 bps. Robust SIP remains competitive in absolute wealth generation: at 50 bps, its final wealth of $21.74 exceeds Nominal CVaR ($20.26) and TC-MinVar ($19.97), though it trails Finite-Regime ($22.52). The $1/N$ portfolio is least sensitive to transaction costs due to its minimal turnover, retaining $28.52 at 50 bps.

5.4 Computational scalability, state density, and bootstrap inference

Figure 10 illustrates the number of active stress states retained by the master problem across all 377 rolling backtest windows. Across the entire 31-year backtest, the algorithm consistently converges while retaining between 1 and 9 active market states (average 3.66) out of the 441 candidate states in $\widehat{\mathcal{U}}$. Under the implemented initialization and one-state-per-iteration update rule, the reported number of active states coincides with the recorded number of master-problem solves (average 3.66 iterations).

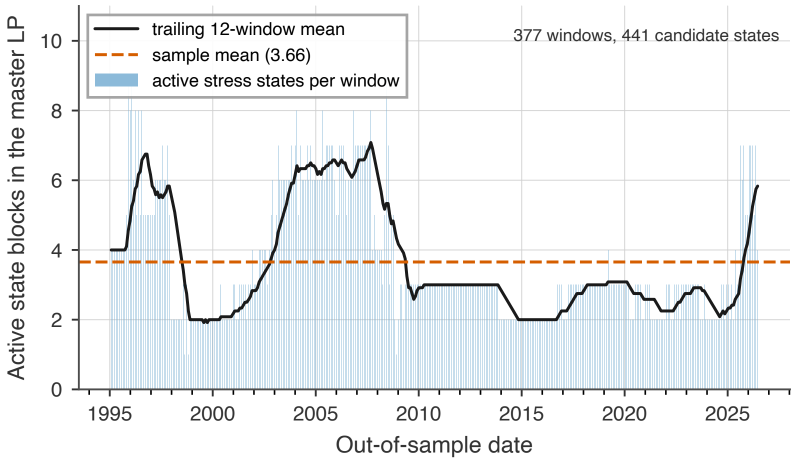

Fig. 10 Active state blocks generated by the grid-restricted constraint-generation algorithm across rolling backtest windows

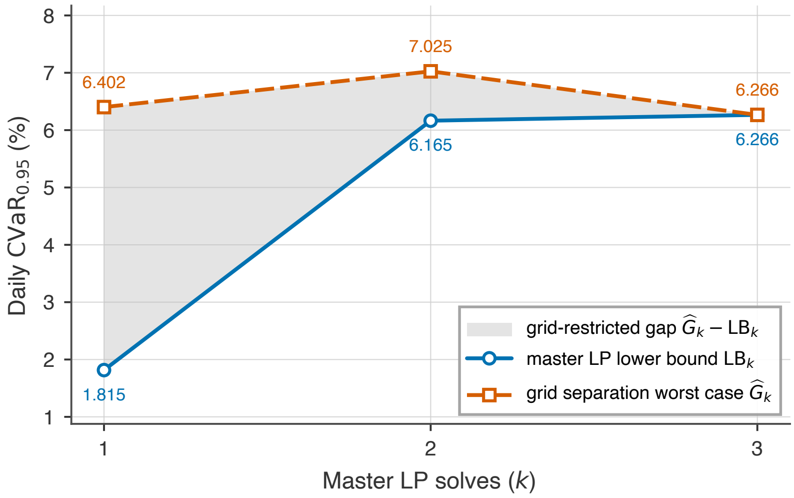

Fig. 11 Grid-restricted convergence of the master-LP lower bound LB_k and full-grid worst-case value $\widehat{G}_k$ for the illustrated rolling window

Figure 11 tracks the convergence of the Master LP Lower Bound (LB_k) and the Grid Separation Worst-Case Value ($\widehat{G}_k$). The shaded green area denotes the grid-restricted exchange gap ($\widehat{G}_k - \mathrm{LB}_k$). Starting from an initial gap at iteration $k = 1$, the oracle worst-case value fluctuates as the portfolio changes, while the master lower bound increases monotonically and the residual gap eventually closes. After 3 exchange iterations, the grid-restricted residual gap falls below 10^{-4}, confirming numerical convergence to the finite robust problem over the 21×21 candidate grid.

Table 7 Computational benchmark: adaptive exchange algorithm vs. dense grid approximation

Method	Status	State blocks	Runtime	Reported value
Adaptive SIP 21×21	Exchange tolerance reached	5	3.45 s	final grid worst-case CVaR $\approx$ 3.3585%
Dense 21×21	OPTIMAL	441	71.43 s	optimum $\approx$ 3.3578%
Dense 41×41	TIME_LIMIT / no primal solution	1681	632.25 s	N/A

Table 8 Geometric grid diagnostic for the 21×21 grid

Diagnostic Metric	Value
Grid-resolution statistic ρ (unit-dependent)	0.032

Table 9 Distribution of the window-level mean ESS across active stress states

Diagnostic Metric	Value
Minimum window-level mean ESS	2.18
5th Percentile window-level mean ESS	16.98
Median window-level mean ESS	72.75
Mean window-level mean ESS	83.02
Maximum window-level mean ESS	264.36
Mean window-level $\tau \times$ ESS tail-support proxy	4.15

Figure 11 illustrates convergence for one selected rolling window. Table 7 reports a single representative-window snapshot used specifically for the dense-grid benchmarking, as running the exact dense 41×41 LP over all 377 windows proved computationally prohibitive. Therefore, the objective values and numbers of active states in Figure 11 and Table 7 are not expected to coincide. For the selected benchmark window and the reported extensive-form implementation, the dense 41×41 LP did not return a primal solution within the 600-second solver limit.

Several diagnostics in Tables 7 and 9 warrant careful interpretation:

1. **Representative-Window ℓ_1 Weight Discrepancy**: On the representative window, the ℓ_1 distance between the adaptive 21×21 exchange solution and the dense 21×21 grid solution is $\|w_{\text{adaptive}} - w_{\text{dense21}}\|_1 \approx 0.00156$. The difference between the adaptive final grid worst-case value and the dense-grid optimum is approximately 7.6×10^{-6}, below the exchange tolerance of 10^{-4}.
2. **Active-State Support**: The minimum, across rolling windows, of the within-window mean ESS over active states is 2.18, indicating that some

windows contain collectively low-support active states. The reported value $\tau \times$ ESS is only a descriptive tail-support proxy and is not a formal tail effective sample size.

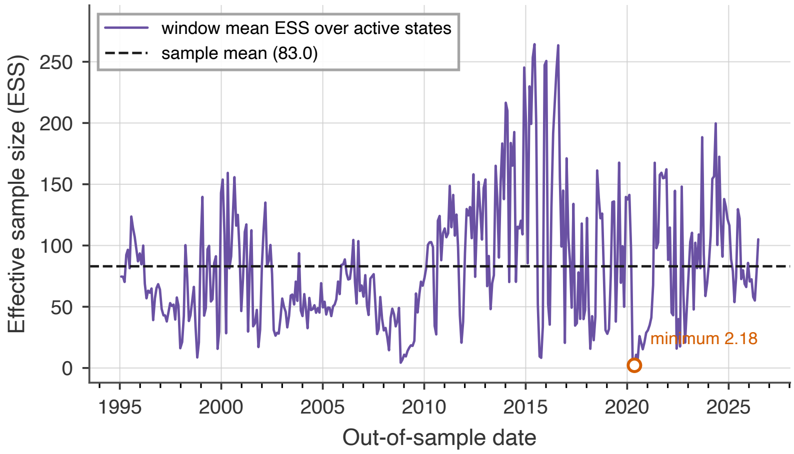

Fig. 12 Evolution of the average Effective Sample Size (ESS) for active stress states across the 377 rolling backtest windows

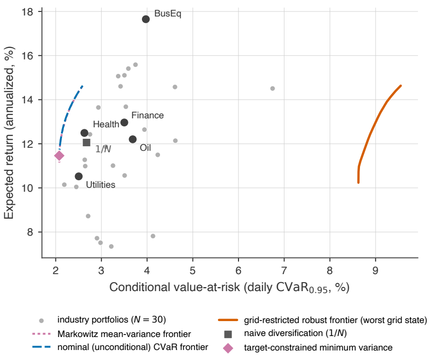

Fig. 13 In-sample risk-return efficient frontiers (15% weight cap)

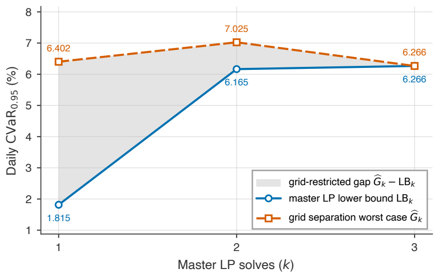

Fig. 14 State density and active stress states. The grid nodes are generated in log-VIX coordinates, and the displayed density includes the Jacobian of the exponential transformation

Figure 13 contextualizes the in-sample decision environment. The grid-restricted robust efficient frontier forms an outer envelope of worst-grid-state CVaR values, positioned to the right of the unconditional nominal CVaR frontier and the classical Markowitz mean-variance frontier. Individual industry portfolios exhibit substantial dispersion in both expected return and downside tail risk, with defensive sectors displaying lower tail risk compared to high-beta sectors.

Figure 14 visualizes the continuous market-state space $\mathcal{U} \subset \mathbb{R}^2$ alongside the bivariate kernel density estimate $f(\text{log-VIX}, \text{Drawdown})$. Historical market states concentrate in a low-volatility, low-drawdown core regime (VIX between 10 and 20, Drawdown below 10%). However, the continuous domain extends outward into severe tail regions, encompassing major historical dislocations such as the 1998 LTCM collapse (VIX 45.74, Drawdown 19.3%), the 2000 to 2002 Dot-Com crash (VIX 45.08, Drawdown 44.7%), the 2008 Lehman Brothers crisis (VIX 80.06, Drawdown 48.4%), the March 2020 COVID-19 shock (VIX 82.69, Drawdown 33.8%), and the 2022 inflation-driven bear market (VIX 34.02, Drawdown 24.5%). In the representative visualization, some active stress states lie in relatively low-density regions of the state domain, exerting binding constraints on the master portfolio problem. To mitigate the computational burden and prevent over-conservatism, one could restrict the Cartesian grid $\widehat{\mathcal{U}}$ to the empirical convex hull of the observed data, thereby eliminating artificial corner states that lack any historical precedent while preserving the interpolative properties of the kernel density.

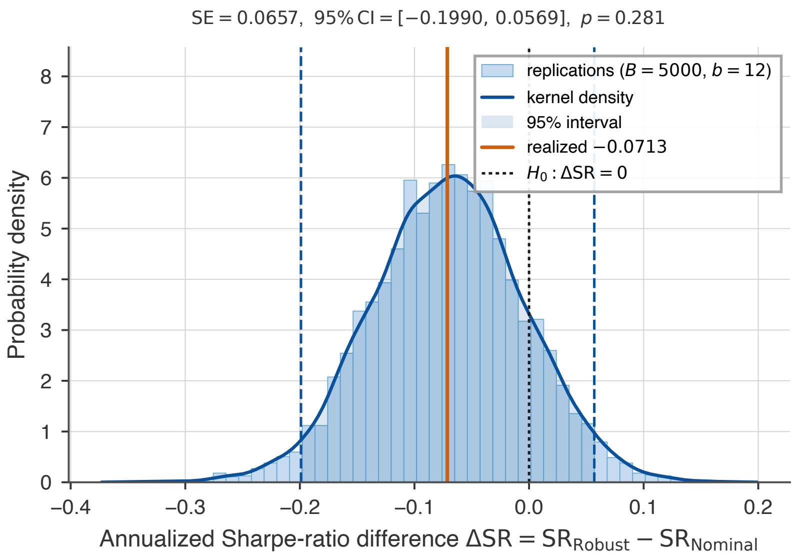

Fig. 15 Paired circular block bootstrap distribution of the annualized Sharpe-ratio difference between Robust SIP and Nominal CVaR

To rigorously assess whether the observed performance differences are statistically distinguishable from sample noise, we implement the circular block percentile bootstrap for the Sharpe-ratio difference. Because financial returns exhibit temporal dependence and volatility clustering, standard i.i.d. t-tests produce severely understated standard errors. This nonparametric inference provides an exploratory assessment of performance stability. We resample holding-period strategy return pairs using a block size of $b = 12$ periods over $B = 5{,}000$ bootstrap replications. Table 10 reports the point estimate, two-sided 95% confidence interval, and bootstrap p-value for the zero-rate Sharpe ratio difference $\Delta\mathrm{SR} = \mathrm{SR}_{\mathrm{Robust}} - \mathrm{SR}_{\mathrm{Benchmark}}$ against all four benchmarks. The bootstrap p-values exceed 0.13 across all benchmarks, indicating that the null hypothesis of equal Sharpe ratios cannot be rejected at conventional significance levels at the 5% level.

Table 10 Paired circular moving-block bootstrap inference for out-of-sample Sharpe ratio differences

Benchmark	ΔSR	Bootstrap SE	95% CI	p-Value
vs. Nominal CVaR	-0.0713	0.0657	[-0.1993, 0.0569]	0.281
vs. 1/N	0.0890	0.0928	[-0.1039, 0.2584]	0.330
vs. TC-MinVar	-0.0643	0.0680	[-0.2133, 0.0552]	0.335
vs. Finite-Regime	-0.0919	0.0613	[-0.2159, 0.0261]	0.130

Against all four benchmarks, the bootstrap p-values exceed 0.05 ($p = 0.281$ vs. Nominal CVaR, $p = 0.330$ vs. $1/N$, $p = 0.335$ vs. TC-MinVar, and $p = 0.130$ vs. Finite-Regime). The bootstrap results therefore do not provide evidence against equality of the Sharpe ratios at the 5% or 10% significance levels. As shown in Figure 15, the bootstrap distribution of ΔSR relative to Nominal CVaR is approximately symmetric and centered near the observed estimate -0.071. The bootstrap confidence intervals provide an exploratory gauge of estimation uncertainty, with the null value $\Delta\mathrm{SR} = 0$ comfortably contained within the 95% confidence interval in all four comparisons.

This statistical finding provides an important nuance to the wealth accumulation results: while Robust SIP generates competitive cumulative wealth (\$25.04 vs. \$22.92 for Nominal CVaR), its risk-adjusted Sharpe ratio is lower due to elevated portfolio volatility. The bootstrap does not provide sufficient evidence to distinguish these Sharpe-ratio differences from zero at conventional significance levels. The primary methodological contribution of the continuous-state framework lies in its principled representation of the state-risk relationship and sparse constraint generation, rather than in statistically dominant out-of-sample risk-adjusted performance.

5.5 Reproducibility, open science, and codebase

To ensure scientific transparency and enable independent verification of all empirical results, the entire computational framework is open-source under an MIT License. The exact code and results corresponding to this manuscript are available in the official release [7].

The repository provides the complete pipeline structured into four main components:

1. **Optimization Engine** (`code/RobustSIP.jl`): Implements the grid-restricted adaptive constraint-generation algorithm, the HiGHS master-LP solver, the full-grid separation oracle, and the benchmark solvers.
2. **Backtesting Suite** (`code/main_exp.jl`): Executes the 31-year rolling-window out-of-sample backtest, computes the 14 performance metrics, performs transaction cost sensitivity sweeps, and conducts the paired circular moving-block bootstrap test.
3. **Empirical Data Module** (`data/aligned_market_data.csv`): Contains the synchronized daily returns for the Kenneth French 30 Industry Portfolios [**?**] alongside CBOE VIX and drawdown indicators over the full 1990 to 2026 period.
4. **Figure Generation Routine** (`code/generate_publication_figures.py`): Reproduces all analytical figures in high-resolution vector PDF format.

Computations were performed on an Apple M1 Pro with 8 CPU cores and 16 GB RAM, using Julia 1.11.6, CSV 0.10.16, DataFrames 1.8.2, JuMP 1.31.1, and HiGHS 1.24.1. Reported solve times encompass model construction and solver execution.

6 Model-specification sensitivity

To ensure the robustness of our findings to methodological hyperparameters, we conduct sensitivity analyses across key design choices: grid resolution, kernel bandwidth, Effective Sample Size (ESS) regularization, and bootstrap block length. The grid resolution and bandwidth analyses (Tables 11 and 12) report averages over a representative subset of 20 evenly spaced rolling out-of-sample windows spanning the full evaluation period from 28 December 1994 to 17 June 2026. This uniform sampling guarantees that the analyses span multiple market regimes, including crisis periods, while remaining computationally tractable for the dense grid benchmarks. The ESS threshold and bootstrap block length sensitivities are evaluated over the entire sequence of 377 rolling windows.

6.1 Grid resolution and bandwidth sensitivity

Table 11 details the sensitivity of the adaptive exchange algorithm to the separation oracle's grid density. The standard implementation uses a 21×21 grid (441 candidate states). Increasing the grid resolution from 11×11 to 81×81 increases the average worst-case CVaR from approximately 4.46% to 4.53%, with limited objective variation beyond the 21×21 grid. Nevertheless, the average ℓ_1 allocation distance between the 21×21 and 81×81 solutions is approximately 0.063, indicating that objective stability does not imply complete allocation stability. Average solve time remains below one second for all examined grids. Table 7 reports one representative-window wall-clock measurement covering kernel-matrix preparation within the solver routine, JuMP model construction, and HiGHS execution. Table 11 reports averages over 20 rolling-window instances. The two timing experiments therefore differ in both selected instances and aggregation and should not be compared as identical benchmarks.

Table 11 Grid size sensitivity: impact on solve time, active states, and objective

Grid Size	Solve Time (s)	Active States	Worst CVaR (%)	Mean ℓ_1 distance to 81×81
11×11	0.20	3.20	4.46	0.1213
21×21	0.26	3.35	4.52	0.0627
41×41	0.47	3.75	4.53	0.0107
81×81	0.76	3.70	4.53	0.0000

Table 12 reports sensitivity to the kernel bandwidth scalar $c \in \{0.5, 0.75, 1.0, 1.5, 2.0\}$. As bandwidth increases, the kernel density estimate becomes smoother and less sensitive to local extreme observations, leading to a reduction in the worst-case CVaR (from 4.60% at $c = 0.5$ to 4.30% at $c = 2.0$). We retain $c = 1.0$ as the prespecified baseline rule-of-thumb bandwidth. The sensitivity results show that smaller bandwidths produce higher worst-case CVaR values, while larger bandwidths smooth local state effects.

Table 12 Bandwidth scalar sensitivity: impact on active states and objective

Bandwidth Multiplier (c)	Active States	Worst CVaR (%)
0.50	3.65	4.60
0.75	3.35	4.56
1.00	3.35	4.52
1.50	3.45	4.38
2.00	3.30	4.30

6.2 ESS regularization and bootstrap block length

To address concerns about boundary scenario overfitting in regions of the state space with low empirical support, we evaluate an ESS-thresholded uncertainty set $\mathcal{U}_{\text{supp}} = \{\theta \in \mathcal{U} : \text{ESS}(\theta) \geq E_{\min}\}$. Table 13 shows that enforcing $E_{\min} = 10$ filters out over half of the candidate states (retaining 172.5 out of 441) without materially altering the worst-case CVaR (4.45%). However, requiring $E_{\min} \geq 40$ more aggressively restricts the admissible state space, which materially reduces the estimated worst-case conditional CVaR by excluding low-support boundary states (worst CVaR drops to 2.69%).

To understand whether this in-sample regularization translates to out-of-sample improvements, we run the complete rolling backtest pipeline across

Table 13 Effective Sample Size (ESS) regularizer sensitivity

ESS Minimum	Retained Grid States	Worst CVaR (%)
0	441.0	4.45
10	172.5	4.45
20	118.8	3.72
40	78.6	2.69
60	61.4	2.52
100	42.4	2.26

four selected ESS thresholds: $E_{\min} \in \{0, 10, 20, 40\}$. Table 14 reports the out-of-sample performance and active state statistics for each case. While $E_{\min} = 10$ successfully excludes over 60% of the grid (retaining an average of 39.3% of states) and excludes candidate grid states whose ESS falls below 10, the point estimates are economically similar, although no formal equality test is conducted. Aggressive regularization ($E_{\min} = 40$) slightly increases the out-of-sample Sharpe ratio but exposes the portfolio to a severe maximum drawdown (-45.94%). We retain $E_{\min} = 0$ as the prespecified primary specification and report $E_{\min} \in \{10, 20, 40\}$ as support-regularized sensitivity analyses.

Table 14 Out-of-sample performance across ESS thresholds ($E_{\min}$)

Threshold	Ann. Ret (%)	Ann. Vol (%)	Sharpe	Max DD (%)	Wealth	Turnover (%)	Avg ESS	Minimum window-level mean active-state ESS	Grid Retained (%)
$E_{\min} = 0$	11.28	13.95	0.81	−41.69	25.04	9.48	83.0	2.2	100.0
$E_{\min} = 10$	11.04	13.92	0.79	−41.72	23.29	9.59	86.9	10.0	39.3
$E_{\min} = 20$	11.02	13.58	0.81	−40.84	23.47	11.13	92.5	20.0	27.5
$E_{\min} = 40$	10.98	13.26	0.83	−45.94	23.54	12.88	112.6	40.0	18.3

Finally, Table 15 verifies the robustness of our statistical inference to the chosen block length b in the paired circular block bootstrap. Across all block lengths from 6 to 24 holding periods, the standard error remains relatively stable, and the two-sided p-value for the Sharpe ratio difference against the Nominal CVaR benchmark never falls below the conventional 0.05 threshold, supporting the qualitative conclusion that the inference is not sensitive to the reported range of block lengths. The block-length sensitivity results use the same methodology as the primary bootstrap analysis. Remaining numerical differences across block lengths reflect both the altered dependence structure induced by the block length and Monte Carlo variability.

Table 15 Bootstrap block length sensitivity on inference

Block Length b, in 21-trading-day holding periods	Bootstrap SE	p-Value
6	0.0655	0.270
9	0.0658	0.275
12	0.0657	0.281
18	0.0618	0.243
24	0.0535	0.183

6.3 Allocation instability and turnover constraints

A notable property of the continuous-state robust formulation is the near-flatness of the worst-case objective surface around the optimum. Because multiple asset combinations can achieve nearly identical worst-case CVaR, the optimal allocation w^* may exhibit instability across adjacent rolling windows. The grid-refinement analysis of Table 11 provides indirect evidence of this flatness: refining the separation grid from 21×21 to 81×81 changes the worst-case CVaR by less than 0.02 percentage points while displacing the optimal allocation by $\|w_{21} - w_{81}\|_1 \approx 0.063$. Objective stability therefore coexists with non-trivial allocation instability, which is consistent with the elevated average holding-period turnover of 9.48% reported in Table 4 (11.37 bps approximate annual transaction-cost drag). While the focus of this paper is unconstrained risk minimization, practical implementations could penalize this instability, for instance by incorporating an ℓ_1 proximity regularizer $\lambda\|w - w_{\text{prev}}\|_1$ into the master LP.

7 Discussion, limitations, and concluding remarks

The proposed state-robust CVaR framework provides a principled and interpretable method for incorporating continuous market covariates into portfolio optimization without imposing artificial discrete regime boundaries. In the reported approximately 31.5-year empirical experiment on US industry portfolios, the continuous-state strategy achieved greater terminal wealth (\$25.04) than TC-MinVar (\$22.16) and nominal CVaR (\$22.92), but lower wealth than naive $1/N$ (\$29.25) and finite-regime CVaR (\$25.68). It also exhibited lower Sharpe ratios and deeper maximum drawdowns than the nominal and finite-regime CVaR benchmarks, and paired circular block percentile bootstrap testing does not provide sufficient evidence to distinguish these Sharpe-ratio differences from zero at conventional significance levels. Thus, the empirical findings do not establish superior risk-adjusted or crisis performance.

The principal contribution of this work is methodological and computational: a kernel-weighted family of conditional return distributions indexed by continuous financial indicators can be incorporated into a continuous-state robust optimization model whose finite grid-restricted approximation can be solved efficiently through adaptive constraint generation. Because the operational implementation uses finite-grid separation ($21 \times 21 = 441$ states), its numerical convergence guarantees apply directly to the grid-restricted robust problem, achieving approximately a $20.7\times$ speedup on the representative benchmark window (single-threaded HiGHS 1.24.1 [**?**], tolerance 10^{-4}) over dense LP discretization while retaining an average of only 3.66 active state blocks (and requiring an average of 3.66 exchange iterations under the implementation's counting convention).

Several methodological considerations and future extensions warrant discussion. First, while the two-dimensional state space (log-VIX and drawdown) remains computationally tractable, kernel density estimation and Cartesian grid search deteriorate rapidly as state dimension increases ($M \geq 3$). Future research should explore dimension-reduction techniques, such as autoencoder latent spaces or conditional generative models, to scale continuous-state robustification. Second, in two dimensions, global grid evaluation provides an efficient separation oracle, whereas in higher dimensions, certified global branch-and-bound or interval arithmetic could provide certified continuous optimality bounds. Third, our empirical tracking of Effective Sample Size revealed that the minimum, across rolling windows, of the within-window mean ESS over active states is 2.18, indicating that some windows contain collectively low-support active stress states. Incorporating explicit ESS-thresholded uncertainty domains $\mathcal{U}_{\text{supp}} = \{\theta \in \mathcal{U} : \text{ESS}(\theta) \geq E_{\min}\}$ represents a natural regularizer to prevent boundary scenario overfitting. Fourth, integrating explicit turnover regularization directly into the master LP would further suppress transaction costs in higher-friction market environments. The framework combines non-parametric conditional risk estimation with a continuous-state semi-infinite formulation, while the empirical implementation solves a finite grid-restricted approximation efficiently through adaptive constraint generation.

Statements and Declarations

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Competing Interests: The authors have no relevant financial or non-financial interests to disclose.

Data Availability: The Kenneth French 30 Industry Portfolio returns [**?**] and CBOE VIX observations [**?**] used in this study were obtained from their publicly accessible source pages. Source URLs, retrieval information, preprocessing code, and the aligned research dataset are documented in the accompanying repository, subject to the original providers' terms of use. All reported results correspond to Zenodo version DOI 10.5281/zenodo.22309074, archived from Git commit 416b0a339040178906642df61a94c25910c11e99. The GitHub repository may contain later development versions [7].

Author Contributions: Madani Bezoui: Conceptualization, Methodology, Software, Writing - original draft. Thiziri Sifaoui: Formal analysis, Validation, Writing - review and editing.

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